

Problem 16.3.15

Evaluate counterclockwise.

$$\oint_C \tan 2\pi z \, dz, \quad C : |z - 0.2| = 0.2$$

Solution

The function $f(z) = \tan 2\pi z = \frac{\sin 2\pi z}{\cos 2\pi z}$ has simple poles at

$$\cos 2\pi z = 0 \implies 2\pi z = \frac{\pi}{2} + n\pi \implies z = \frac{1}{4} + \frac{n}{2}, \quad n \in \mathbb{Z}$$

Only $z_0 = \frac{1}{4}$ lies in C ,

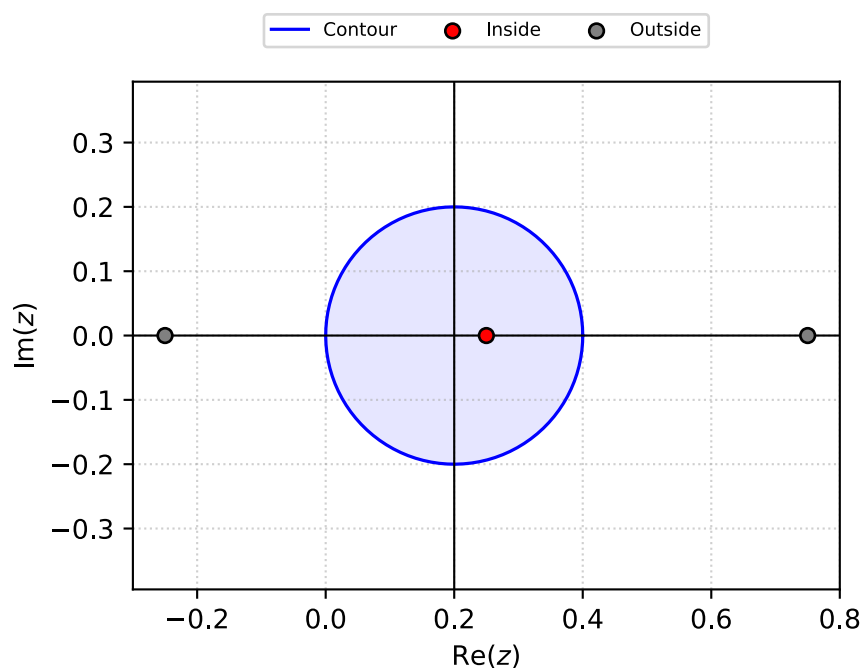


Figure 1: Poles

$$\operatorname{Res}_{z=z_0} f(z) = \frac{p(z_0)}{q'(z_0)} = \frac{\sin 2\pi z}{-2\pi \sin 2\pi z} = -\frac{1}{2\pi}$$

The integral is then in the form,

$$\oint_C f(z) \, dz = 2\pi i \operatorname{Res}_{z=z_0} f(z)$$

$$\oint_C \tan 2\pi z \, dz = 2\pi i \operatorname{Res}_{z=\frac{1}{4}} f(z) = \boxed{-i}$$

Problem 16.3.17

Evaluate counterclockwise.

$$\oint_C \frac{e^z}{\cos z} dz, \quad C : \left| z - \frac{\pi i}{2} \right| = 4.5$$

Solution

The function $f(z) = \frac{e^z}{\cos z}$ has simple poles at

$$\cos z = 0 \implies z = \frac{\pi}{2} + n\pi, \quad n \in \mathbb{Z}$$

Poles at $z_{1,2} = \pm \frac{\pi}{2}$ lie within the region,

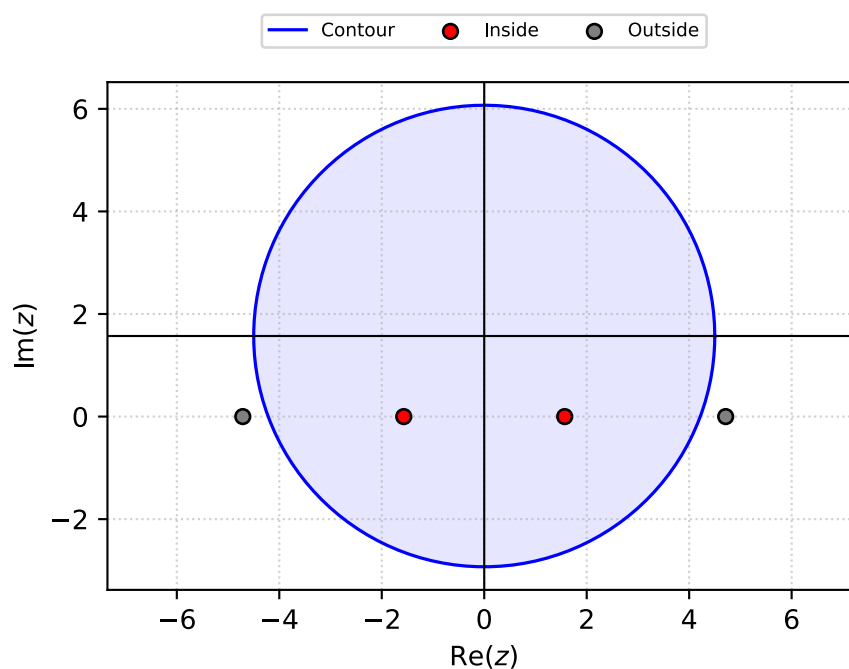


Figure 2: Poles

$$\operatorname{Res}_{z=z_0} f(z) = \frac{p(z_0)}{q'(z_0)} = -\frac{e^{z_0}}{\sin z_0}$$

$$\operatorname{Res}_{z=\frac{\pi}{2}} f(z) = -\frac{e^{\frac{\pi}{2}}}{\sin \frac{\pi}{2}} = -e^{\frac{\pi}{2}}$$

$$\operatorname{Res}_{z=-\frac{\pi}{2}} f(z) = -\frac{e^{-\frac{\pi}{2}}}{\sin -\frac{\pi}{2}} = e^{-\frac{\pi}{2}}$$

The integral is then in the form,

$$\oint_C f(z) dz = 2\pi i \sum_{j=1}^k \operatorname{Res}_{z=z_j} f(z)$$

$$\oint_C \frac{e^z}{\cos z} dz = 2\pi i \left(-e^{\frac{\pi}{2}} + e^{-\frac{\pi}{2}} \right) = \boxed{-4\pi i \sinh\left(\frac{\pi}{2}\right)}$$

Problem 16.3.19

Evaluate counterclockwise.

$$\oint_C \frac{\sinh z}{2z - i} dz, \quad C : |z - 2i| = 2$$

Solution

The function $f(z) = \frac{\sinh z}{2z - i}$ has simple poles at

$$2z - i = 0 \implies z = \frac{i}{2}$$

The pole at $z_0 = \frac{i}{2}$ lies within the region,

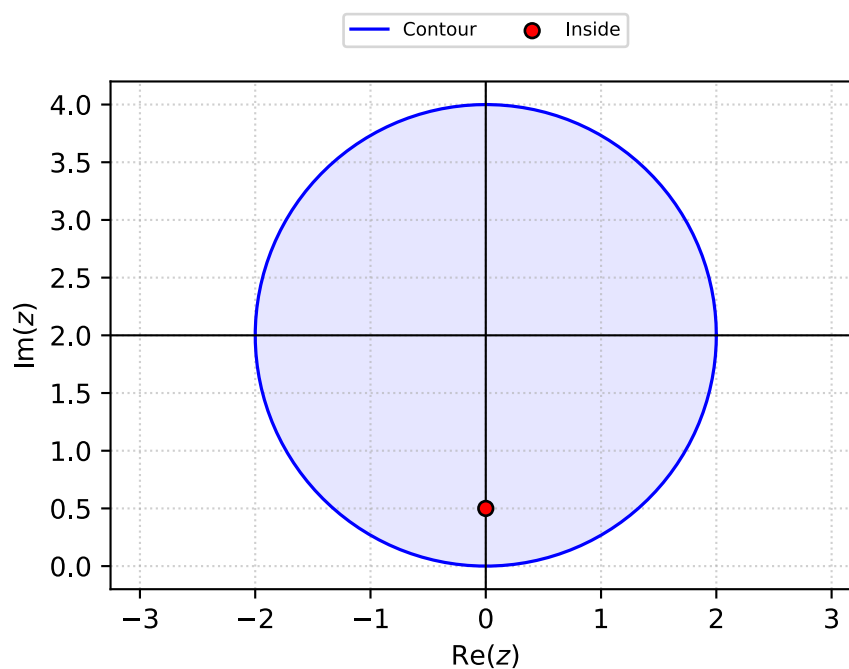


Figure 3: Poles

$$\operatorname{Res}_{z=z_0} f(z) = \frac{p(z_0)}{q'(z_0)} = \frac{1}{2} \sinh z_0$$

$$\operatorname{Res}_{z=\frac{i}{2}} f(z) = \frac{1}{2} \sinh \frac{i}{2} = \frac{i}{2} \sin \frac{1}{2}$$

The integral is then in the form,

$$\oint_C f(z) dz = 2\pi i \operatorname{Res}_{z=z_0} f(z)$$

$$\oint_C \frac{\sinh z}{2z - i} dz = 2\pi i \left(\frac{i}{2} \sin \frac{1}{2} \right) = \boxed{-\pi \sin \frac{i}{2}}$$

Problem 16.3.21

Evaluate counterclockwise.

$$\oint_C \frac{\cos \pi z}{z^5} dz, \quad C : |z| = \frac{1}{2}$$

Solution

The function $f(z) = \frac{\cos \pi z}{z^5}$ has a 5th-order pole at

$$z^5 = 0 \implies z = 0$$

The pole at $z_0 = 0$ lies within the region,

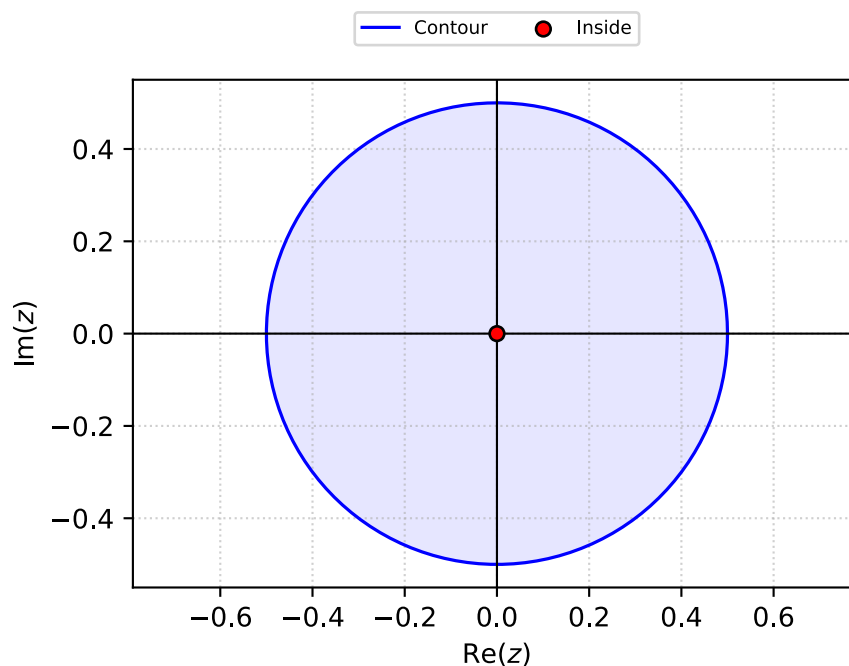


Figure 4: Poles

$$\operatorname{Res}_{z=0} f(z) = \frac{1}{4!} \lim_{z \rightarrow 0} \left\{ \frac{d^4}{dz^4} [z^4 f(z)] \right\} = \frac{1}{24} \frac{d^4}{dz^4} [\cos \pi z] \Big|_{z=0} = \frac{\pi^4}{24} \cos \pi z \Big|_{z=0} = \frac{\pi^4}{24}$$

The integral is then in the form,

$$\oint_C f(z) dz = 2\pi i \operatorname{Res}_{z=z_0} f(z)$$

$$\oint_C \frac{\cos \pi z}{z^5} dz = 2\pi i \left(\frac{\pi^4}{24} \right) = \boxed{\frac{i\pi^5}{12}}$$

Problem 16.4.1

Evaluate.

$$\int_0^\pi \frac{2}{k - \cos \theta} d\theta$$

Solution

The integral can be converted to a complex integral by substituting $z = e^{i\theta}$

$$\int_0^\pi \frac{2}{k - \cos \theta} d\theta = \int_0^{2\pi} \frac{1}{k - \cos \theta} d\theta = \oint_C \frac{1}{k - \frac{1}{2}(z + z^{-1})} \frac{dz}{iz} = 2i \oint_C \frac{1}{z^2 - 2kz + 1} dz$$

Where C is the unit circle in the complex plane.

$f(z) = \frac{1}{z^2 - 2kz + 1}$ has simple poles where

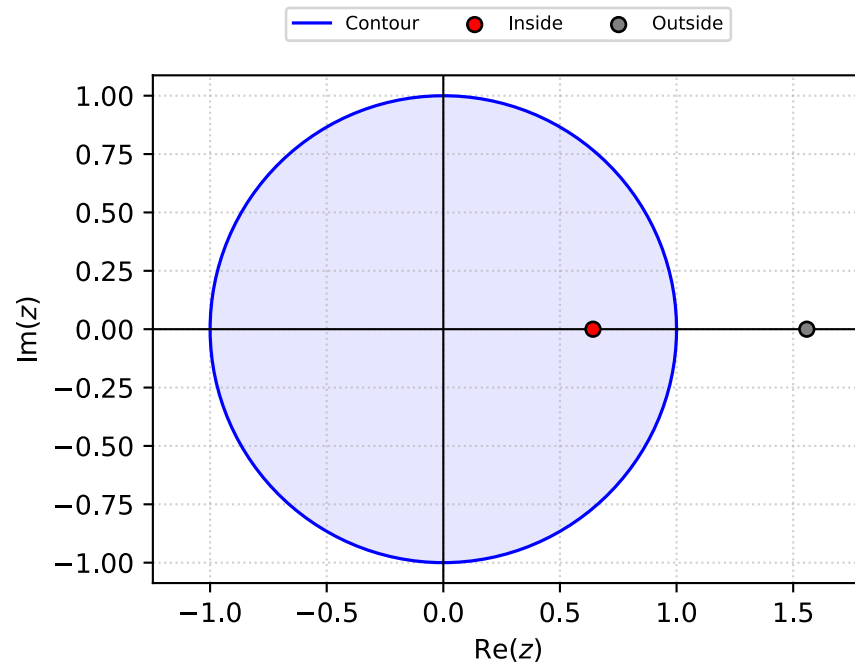
$$z^2 - 2kz + 1 = 0 \implies z = \frac{2k \pm \sqrt{4k^2 - 4}}{2} = k \pm \sqrt{k^2 - 1}$$

$$z_1 = k - \sqrt{k^2 - 1} \quad z_2 = k + \sqrt{k^2 - 1}$$

$$\operatorname{Res}_{z=z_0} f(z) = \frac{1}{2z_0 - 2k} = \frac{1}{2(z_0 - k)}$$

$$\operatorname{Res}_{z=z_{1,2}} f(z) = \frac{1}{2(k \pm \sqrt{k^2 - 1} - k)} = \pm \frac{1}{2\sqrt{k^2 - 1}}$$

Assuming $k > 1$, only z_1 lies within C ,

Figure 5: Poles (e.g. $k = 1.1$)

$$\begin{aligned}
 \oint_C f(z) \, dz &= 2\pi i \operatorname{Res}_{z=z_0} f(z) \\
 \oint_C \frac{1}{z^2 - 2kz + 1} &= 2\pi i \left(-\frac{1}{2\sqrt{k^2 - 1}} \right) = -\frac{\pi i}{\sqrt{k^2 - 1}} \\
 \int_0^\pi \frac{2}{k - \cos \theta} \, d\theta &= 2i \left(-\frac{\pi i}{\sqrt{k^2 - 1}} \right) \\
 &= \boxed{\frac{2\pi}{\sqrt{k^2 - 1}}}
 \end{aligned}$$

Problem 16.4.7

Evaluate.

$$\int_0^{2\pi} \frac{a}{a - \sin \theta} d\theta$$

Solution

The integral can be converted to a complex integral by substituting $z = e^{i\theta}$

$$\int_0^{2\pi} \frac{a}{a - \sin \theta} d\theta = \oint_C \frac{a}{a - \frac{1}{2i}(z - z^{-1})} \frac{dz}{iz} = -2a \oint_C \frac{1}{z^2 - 2aiz - 1} dz$$

Where C is the unit circle in the complex plane.

$f(z) = \frac{1}{z^2 - 2aiz - 1}$ has simple poles where

$$z^2 - 2aiz - 1 = 0 \implies z = \frac{2ai \pm \sqrt{-4a^2 + 4}}{2} = ai \pm \sqrt{1 - a^2}$$

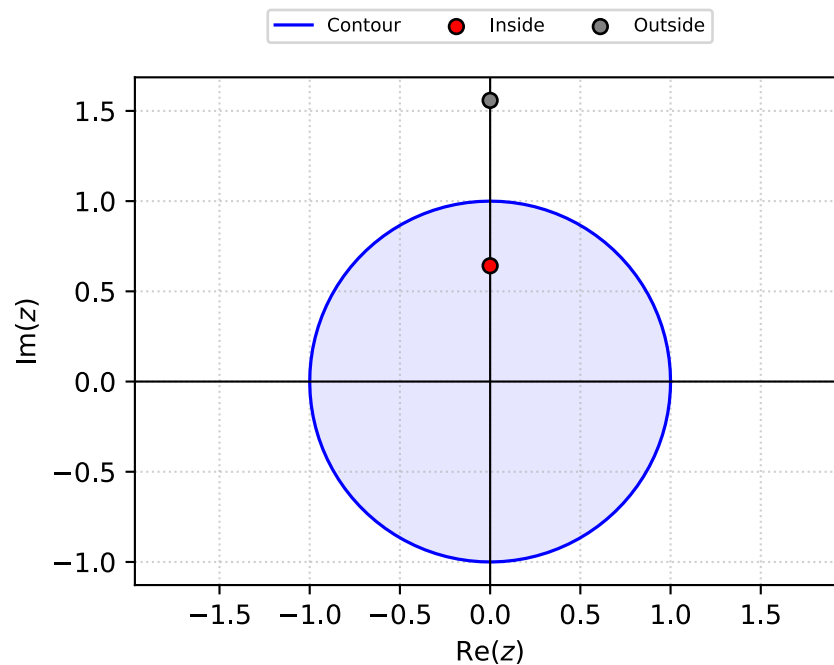
Assuming $a > 1$

$$z_1 = (a - \sqrt{a^2 - 1})i \quad z_2 = (a + \sqrt{a^2 - 1})i$$

$$\operatorname{Res}_{z=z_0} f(z) = \frac{1}{2z_0 - 2ai} = \frac{1}{2(z_0 - ai)}$$

$$\operatorname{Res}_{z=z_{1,2}} f(z) = \frac{1}{2((a \pm \sqrt{a^2 - 1})i - ai)} = \pm \frac{1}{2i\sqrt{a^2 - 1}}$$

Only z_1 lies within C ,

Figure 6: Poles (e.g. $a = 1.1$)

$$\begin{aligned}
 \oint_C f(z) \, dz &= 2\pi i \operatorname{Res}_{z=z_0} f(z) \\
 \oint_C \frac{1}{z^2 - 2kz + 1} &= 2\pi i \left(-\frac{1}{2i\sqrt{a^2 - 1}} \right) = -\frac{\pi}{\sqrt{a^2 - 1}} \\
 \int_0^{2\pi} \frac{a}{a - \sin \theta} \, d\theta &= -2a \left(-\frac{\pi}{\sqrt{a^2 - 1}} \right) \\
 &= \boxed{\frac{2a\pi}{\sqrt{a^2 - 1}}}
 \end{aligned}$$

Problem 16.4.9

Evaluate.

$$\int_0^{2\pi} \frac{\cos \theta}{13 - 12 \cos 2\theta} d\theta$$

Solution

The integral can be converted to a complex integral by substituting $z = e^{i\theta}$

$$\int_0^{2\pi} \frac{\cos \theta}{13 - 12 \cos 2\theta} d\theta = \oint_C \frac{\frac{1}{2}(z + z^{-1})}{13 - 12\left(\frac{1}{2}(z^2 + z^{-2})\right)} \frac{dz}{iz} = \oint_C \frac{z^2 + 1}{2z} \cdot \frac{z^2}{-6z^4 + 13z^2 - 6} \cdot \frac{dz}{iz} = \frac{i}{2} \oint_C \frac{z(z^2 + 1)}{6z^4 - 13z^2 + 6} dz$$

Where C is the unit circle in the complex plane.

$f(z) = \frac{z(z^2 + 1)}{6z^4 - 13z^2 + 6}$ has simple poles where

$$6z^4 - 13z^2 + 6 = 0 \implies z^2 = \frac{13 \pm \sqrt{169 - 144}}{6} = \frac{13 \pm 5}{12} \implies z = \pm \sqrt{\frac{2}{3}}, \pm \sqrt{\frac{3}{2}}$$

$$z_1 = -\sqrt{\frac{2}{3}} \quad z_2 = \sqrt{\frac{2}{3}} \quad z_3 = -\sqrt{\frac{3}{2}} \quad z_4 = \sqrt{\frac{3}{2}}$$

Only z_1 and z_2 lie within C ,

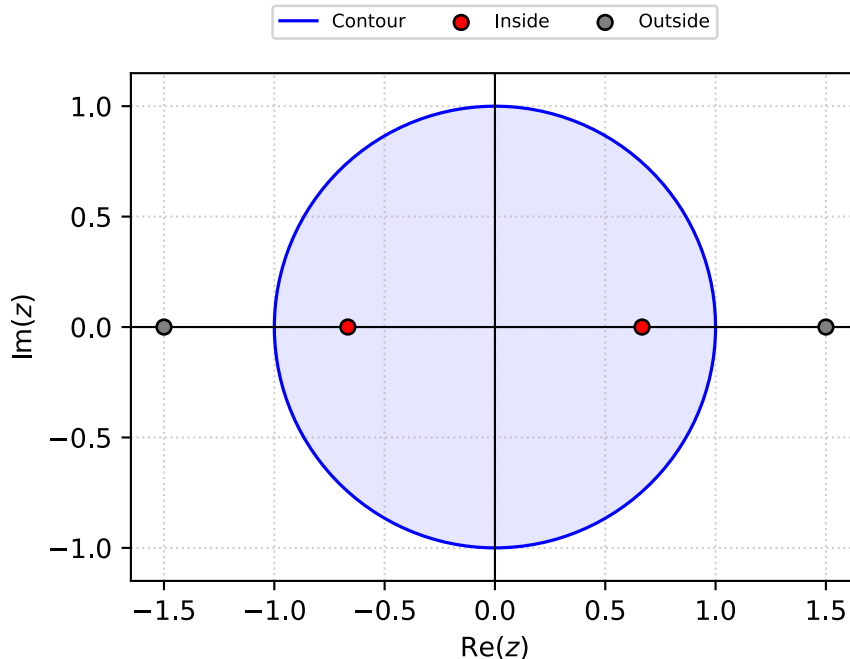


Figure 7: Poles

$$\text{Res}_{z=z_0} f(z) = \frac{z_0(z_0^2 + 1)}{24z_0^3 - 26z_0} = \frac{z_0^2 + 1}{24z_0^2 - 26}$$

$$\operatorname{Res}_{z=z_{1,2}} f(z) = \frac{\left(\pm\sqrt{\frac{2}{3}}\right)^2 + 1}{24\left(\pm\sqrt{\frac{2}{3}}\right)^2 - 26} = -\frac{1}{6}$$

The integral is then in the form,

$$\oint_C f(z) \, dz = 2\pi i \sum_{j=1}^k \operatorname{Res}_{z=z_j} f(z)$$

$$\oint_C \frac{z(z^2 + 1)}{6z^4 - 13z^2 + 6} = 2\pi i \left(-\frac{1}{6} - \frac{1}{6}\right) = -\frac{2\pi i}{3}$$

$$\int_0^{2\pi} \frac{\cos \theta}{13 - 12 \cos 2\theta} \, d\theta = \frac{i}{2} \left(-\frac{2\pi i}{3}\right) = \boxed{\frac{\pi}{3}}$$

Problem 16.4.11

Evaluate.

$$\int_{-\infty}^{\infty} \frac{1}{(1+x^2)^2} dx$$

Solution

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum \text{Res } f(z)$$

$f(z) = \frac{1}{(1+z^2)^2}$ has second order poles where

$$(1+z^2)^2 = 0 \implies z = \pm i$$

Only $z = i$ lies in the upper half contour.

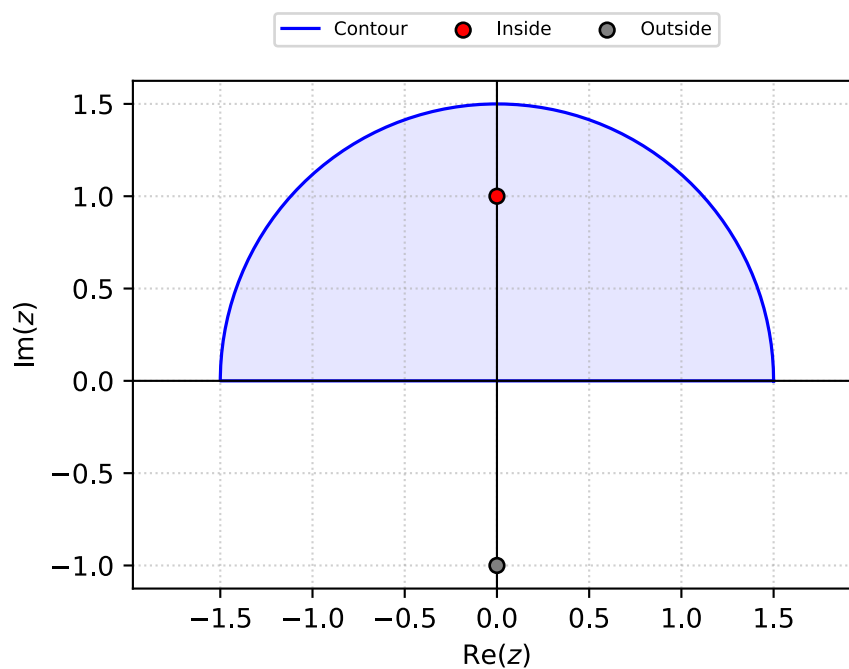


Figure 8: Poles

$$\begin{aligned}\operatorname{Res}_{z=i} f(z) &= \lim_{z \rightarrow i} \left\{ \frac{d}{dz} \left[\frac{(z-i)^2}{(1+z^2)^2} \right] \right\} \\ &= \lim_{z \rightarrow i} \left\{ \frac{d}{dz} \left[\frac{(z-i)^2}{(z-i)^2 (z+i)^2} \right] \right\} \\ &= \lim_{z \rightarrow i} \frac{d}{dz} \frac{1}{(z+i)^2} \\ &= \lim_{z \rightarrow i} \frac{-2}{(z+i)^3} \\ &= -\frac{i}{4}\end{aligned}$$

$$\int_{-\infty}^{\infty} \frac{1}{(1+x^2)^2} dx = 2\pi i \left(-\frac{i}{4} \right) = \boxed{\frac{\pi}{2}}$$

Problem 16.4.15

Evaluate.

$$\int_{-\infty}^{\infty} \frac{x^2}{x^6 + 1} dx$$

Solution

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum \text{Res } f(z)$$

$f(z) = \frac{z^2}{z^6+1}$ has simple poles where

$$z^6 + 1 = 0 \implies z_n = \exp\left\{\frac{i\pi}{6}(1 + 2n)\right\}, \text{ for } n = 0, 1, 2, 3, 4, 5$$

Only z_1, z_2 , and z_3 lie in the upper half contour.

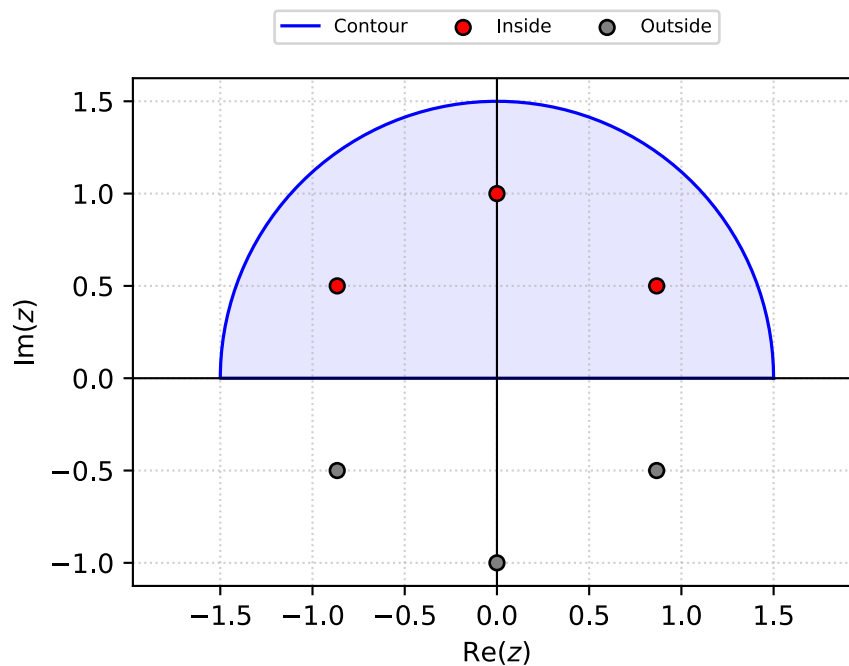


Figure 9: Poles

$$\text{Res}_{z=z_0} f(z) = \frac{z_0^2}{6z_0^5} = \frac{1}{6z_0^3}$$

$$\text{Res}_{z=z_n} f(z) = \frac{1}{6 \exp\left\{\frac{i\pi}{6}(1 + 2n)\right\}^3} = \frac{1}{6} \exp\left\{-\frac{i\pi}{2}(1 + 2n)\right\}$$

$$\text{Res}_{z=z_0} f(z) = -\frac{i}{6} \quad \text{Res}_{z=z_1} f(z) = \frac{i}{6} \quad \text{Res}_{z=z_2} f(z) = -\frac{i}{6}$$

$$\int_{-\infty}^{\infty} \frac{x^2}{x^6 + 1} dx = 2\pi i \left(-\frac{i}{6} + \frac{i}{6} - \frac{i}{6} \right) = 2\pi i \left(-\frac{i}{6} \right) = \boxed{\frac{\pi}{3}}$$

Problem 16.4.17

Evaluate.

$$\int_{-\infty}^{\infty} \frac{\sin 3x}{x^4 + 1} dx$$

Solution

$$\int_{-\infty}^{\infty} \frac{\sin 3x}{x^4 + 1} dx = \int_{-\infty}^{\infty} \frac{ODD}{EVEN} = \int_{-\infty}^{\infty} ODD = \boxed{0}$$

Problem 16.4.19

Evaluate.

$$\int_{-\infty}^{\infty} \frac{1}{x^4 - 1} dx$$

Solution

$$\int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum_{\text{upper half}} \text{Res } f(z) + \pi i \sum_{z_0 \in \mathbb{R}} \text{Res } f(z)$$

$f(z) = \frac{1}{z^4 - 1}$ has simple poles where

$$z^4 - 1 = 0 \implies z_n = \pm 1, \pm i$$

$z_{1,2} = \pm 1$ lies on the real axis, $z_3 = i$ lies in the upper half plane.

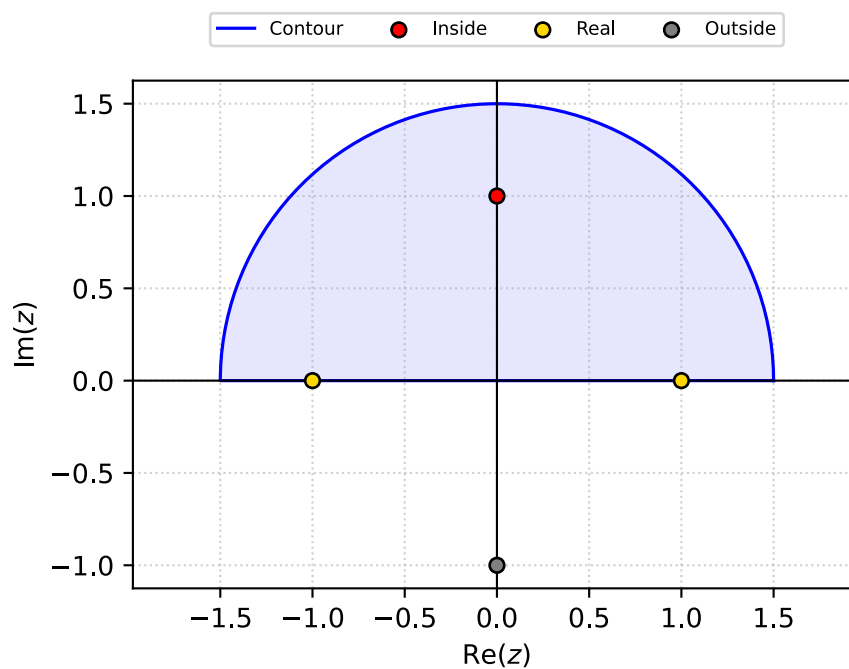


Figure 10: Poles

$$\text{Res}_{z=z_0} f(z) = \frac{1}{4z_0^3}$$

$$\text{Res}_{z=1} f(z) = \frac{1}{4} \quad \text{Res}_{z=-1} f(z) = -\frac{1}{4} \quad \text{Res}_{z=i} f(z) = \frac{i}{4}$$

$$\int_{-\infty}^{\infty} \frac{x^2}{x^6 + 1} dx = 2\pi i \left(\frac{i}{4} \right) + \pi i \left(\frac{1}{4} - \frac{1}{4} \right) = \boxed{-\frac{\pi}{2}}$$

Problem 16.4.21

Evaluate.

$$\int_{-\infty}^{\infty} \frac{\sin x}{(x-1)(x^2+4)} dx$$

Solution

$$\int_{-\infty}^{\infty} f(x) \sin x dx = -2\pi \sum \text{Im Res} [f(z)e^{iz}]$$

$f(z)e^{iz} = \frac{e^{iz}}{(z-1)(z^2+4)}$ has simple poles where

$$z - 1 = 0 \implies z_1 = 1$$

$$z^2 + 4 = 0 \implies z_{2,3} = \pm 2i$$

$z_1 = 1$ lies on the real axis, $z_2 = 2i$ lies in the upper half plane.

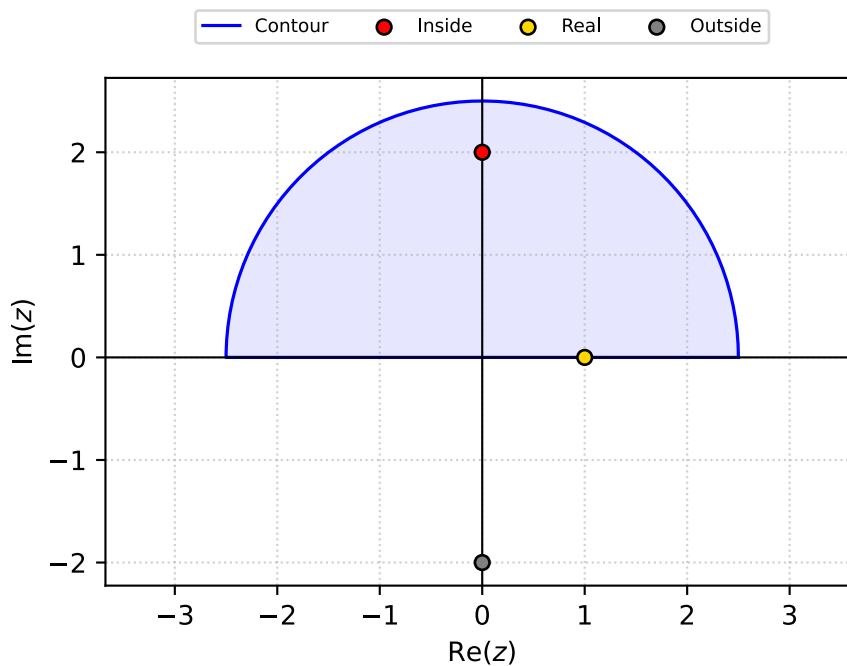


Figure 11: Poles

$$\text{Res}_{z=z_0} f(z) = \frac{e^{iz_0}}{((z_0-1)(z_0^2+4))'} = \frac{e^{iz_0}}{(z_0^3 - z_0^2 + 4z_0 - 4)'} = \frac{e^{iz_0}}{3z_0^2 - 2z_0 + 4}$$

$$\text{Res}_{z=1} f(z) = \frac{e^i}{5}$$

$$\text{Res}_{z=2i} f(z) = \frac{e^{-2}}{-4(2+i)} = \frac{e^{-2}(i-2)}{20}$$

$$\begin{aligned}
\int_{-\infty}^{\infty} f(z)e^{iz} dz &= 2\pi i \left(\frac{e^{-2}(i-2)}{20} \right) + \pi i \left(\frac{e^i}{5} \right) \\
&= \frac{\pi i e^{-2}(i-2)}{10} + \frac{\pi i e^i}{5} \\
&= \frac{\pi}{10} [(-e^{-2} - 2ie^{-2}) + 2(\cos 1 + i \sin 1)] \\
&= \frac{\pi}{10} [(-e^{-2} - 2ie^{-2}) + 2i(\cos 1 + i \sin 1)] \\
&= \frac{\pi}{10} [-e^{-2} - 2ie^{-2} + 2i \cos 1 - 2 \sin 1] \\
&= \frac{\pi}{10} [-(e^{-2} + 2 \sin 1) + 2i(\cos 1 - e^{-2})]
\end{aligned}$$

The principal value of the original integral is the imaginary part

$$\text{pr. v. } \int_{-\infty}^{\infty} \frac{\sin x}{(x-1)(x^2+4)} dx = \boxed{\frac{\pi}{5} (\cos 1 - e^{-2})}$$

Problem 16.4.25

Find the Cauchy principal value.

$$\int_{-\infty}^{\infty} \frac{x+5}{x^3-x} dx$$

Solution

$f(z) = \frac{z+5}{z^3-z}$ has simple poles where

$$z^3 - z = 0 \implies z(z+1)(z-1) = 0 \implies \begin{cases} z_1 = -1 \\ z_2 = 0 \\ z_3 = 1 \end{cases}$$

All poles lie on the real axis

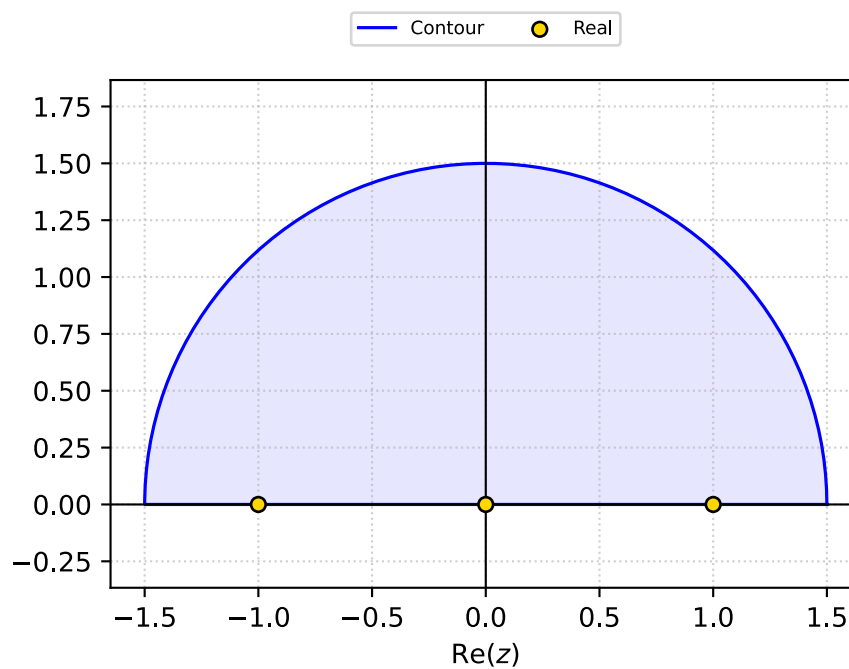


Figure 12: Poles

$$\int_{-\infty}^{\infty} f(x) dx = \pi i \sum_{z_0 \in \mathbb{R}} \text{Res } f(z)$$

$$\text{Res}_{z=z_0} f(z) = \frac{z_0 + 5}{3z_0^2 - 1}$$

$$\text{Res}_{z=-1} f(z) = 2$$

$$\text{Res}_{z=0} f(z) = -5$$

$$\operatorname{Res}_{z=1} f(z) = 3$$

$$\int_{-\infty}^{\infty} \frac{x+5}{x^3-x} dx = \pi i (2-5+3) = \boxed{0}$$